

KIRAN WILLIAMS

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Professional Summary

Data and business analytics professional with more than eight years of experience across operations, education, customer service, and workforce coordination. Currently pursuing an M.S. in Data Analytics and an MBA, with hands-on project experience using SQL, Python, Excel, and Tableau to clean, analyze, visualize, and interpret data. Applied statistical analysis, machine learning, dashboard development, and data management to investigate business, operational, and academic problems. Combines analytical thinking, process-improvement experience, and stakeholder communication to translate complex findings into practical recommendations.

Technical Skills

Programming & Libraries	SQL, Python, R, Pandas, NumPy, Scikit-learn, Statsmodels, Matplotlib, Seaborn
Tools	Excel, Tableau, Google Sheets, Cognos Analytics, MySQL Workbench, Jupyter Notebook, GitHub
Data Analysis	Exploratory data analysis, data cleaning, statistical analysis, data visualization, feature engineering, KPI development, correlation analysis
Machine Learning	Classification, logistic regression, Naive Bayes, decision trees, k-means clustering, PCA, anomaly detection, model evaluation
Data Management	ETL, data validation, relational databases, schema design, CTEs, window functions, data governance
Business Analysis	Requirements definition, operational reporting, root-cause analysis, process improvement, stakeholder communication, decision support

Experience

Retail Operations Associate
Kohl's, Laurel, MD

Nov 2025-Present

- Fulfill and package 50–100 customer orders weekly while supporting the receipt and unloading of 2,000–8,000 merchandise units per truck.
- Assist customers with Amazon returns and troubleshoot technology and transaction issues to complete returns accurately and reduce service delays.
- Generated nine Kohl's Rewards and 12 Sephora Beauty Insider enrollments during first shift at the Amazon returns counter, contributing to recognition as Associate of the Month within four months.

Guest Services & Operations Supervisor
Chesapeake Employers Insurance Arena, Baltimore, MD

Dec 2018-Present

- Supervise 15–20 employees during 20–30 annual events averaging approximately 5,000 attendees, adjusting assignments to maintain guest-service and operational coverage.
- Trained more than 50 employees on service, safety, and event procedures while communicating staffing and operational recommendations to senior leadership.
- Coordinate parking assistance and troubleshoot guest technology issues during high-volume events, helping maintain an organized arrival experience.

Middle School Science Teacher
Anne Arundel County Public Schools, Odenton, MD

Aug 2023-Jun 2025

- Analyzed academic performance, attendance, and behavior data for more than 300 students in grades 6–8 to identify learning needs and coordinate targeted support.
- Integrated literacy strategies into science instruction and monitored assessment data, supporting one-grade-level reading gains among approximately 25% of students taught.
- Served as sixth-grade science department chair for six months, facilitating planning meetings, coordinating instructional priorities, and supporting curriculum alignment.

Workforce Specialist
UMBC Facilities Management, Baltimore, MD

Oct 2021-Aug 2023

- Tracked work-order response times against baseline, six-month, and one-year results, helping identify process changes associated with a 20% reduction in response time.
- Organized work-order assignments and employee schedules using Google Calendar and analyzed Google Forms survey responses to support staffing and process decisions.
- Coordinated recruitment, onboarding, scheduling, and training for 20–30 student employees, supporting consistent facilities staffing and workforce coverage.

High School Science Teacher
Prince George's County Public Schools, Hyattsville, MD

Sep 2020-Jun 2021

- Developed recurring Excel reports to monitor academic performance and attendance for 150–200 students in grades 9–10 during COVID-era instruction.
- Used performance data to group students for differentiated instruction and identify individuals requiring additional academic or attendance support.
- Communicated performance and attendance findings to students, families, and school stakeholders to coordinate follow-up and instructional adjustments.

Relevant Projects

Identifying Citi Bike Availability Pressure Through Trip Patterns

[GitHub Project](#)

- **Problem:** Citi Bike trip demand changes throughout the day and across the station network, but total monthly activity can hide time-specific differences between bike pickups and returns. Operations teams need a defensible way to identify stations and time periods that may require closer monitoring without treating completed-trip data as proof that a station was empty or full.
- **Process:** Combined five May 2026 source files into a MySQL staging table and validated 4,674,903 records for completeness, unique ride identifiers, timestamp order, categorical consistency, station mapping, coordinates, and source reconciliation. Converted text timestamps to millisecond-precision datetime fields, created reusable time and duration dimensions, preserved source lineage, and flagged 478 reporting-boundary rides and 22 rides lasting longer than 24 hours. Used SQL to analyze demand by date and hour, rider and bike segments, ride duration, station activity, and directional station flow.
- **Tools:** MySQL, MySQL Workbench, SQL, ETL, CTEs, conditional aggregation, data validation, business-rule implementation, and analytical indexing.
- **Solution:** Created a reproducible analytical table and SQL workflow that measured departures, arrivals, total station activity, net flow, absolute imbalance, and time-specific directional patterns. Developed documented findings and an operational monitoring framework that prioritizes station-hour combinations for further investigation while clearly distinguishing trip-flow proxies from confirmed availability failures.
- **Outcome:** Reconciled all 4,674,903 rides through a final PASS quality gate and found that monthly station totals could conceal substantial hourly reversals. At 9 Ave & W 33 St, the analysis identified 1,324 net arrivals at 8 a.m. and 1,705 net departures at 5 p.m., supporting a recommendation for time-specific monitoring using live inventory, dock-capacity, outage, and rebalancing data.
- **Key Skills:** SQL analysis, ETL, data cleaning, validation, schema design, feature engineering, time-series aggregation, station-flow analysis, data governance, operational decision support, and executive recommendations.

Identifying Academic Risk Through Integrated Student Data

[GitHub Project](#)

- **Problem:** Student information was distributed across grades, attendance, behavior, and intervention records. Reviewing these measures separately made it difficult to identify patterns, compare student groups, and determine which students might require earlier follow-up.
- **Process:** Integrated data for 120 students and created calculated fields to measure academic performance, attendance patterns, behavior, and intervention activity. Used parameters and level-of-detail expressions to support comparisons across the school, student subgroups, and individual students. Analyzed the relationship between attendance and grades and developed interactive dashboard actions for further investigation.
- **Tools:** Tableau, calculated fields, parameters, level-of-detail expressions, filters, dashboard actions, and data validation.
- **Solution:** Developed three connected Tableau dashboards: a schoolwide performance overview, a subgroup analysis view, and an individual student profile. The dashboards allowed users to move from high-level indicators to detailed student records while keeping intervention decisions under human review.
- **Outcome:** Identified a 0.507 correlation between attendance and grades and found that students with attendance below 85% had average grades 12.37 points lower than other students. The analysis provided a structured way for school leaders to identify students for earlier review and investigate the factors contributing to academic risk.
- **Key Skills:** Data integration, dashboard development, calculated metrics, segmentation, correlation analysis, requirements translation, data storytelling, and responsible use of student data.

Balancing Revenue and Unit Demand for Product Strategy

[GitHub Project](#)

- **Problem:** The transaction data did not clearly distinguish between products that generated the most revenue and products that generated the greatest unit demand. This made it difficult to evaluate product performance or compare sales patterns across categories and cities.
- **Process:** Cleaned 30,394 raw rows and produced 30,206 validated transaction lines by checking data quality and removing unusable records. Used PivotTables to summarize revenue, orders, units sold, product categories, and geographic performance. Compared product rankings based on revenue and unit volume to identify differences between financial value and customer demand.
- **Tools:** Microsoft Excel, PivotTables, PivotCharts, slicers, data-cleaning functions, calculated KPIs, and interactive dashboard design.
- **Solution:** Built an interactive Excel dashboard covering \$5.64 million in revenue, 29,018 orders, and 33,969 units sold. Added revenue, order, and unit KPIs along with category and city slicers so users could compare product and market performance without rebuilding the analysis.
- **Outcome:** Found that laptops and phones generated 61% of total revenue, while batteries led unit sales but contributed less than 1% of revenue. The analysis separated value-driving products from demand-driving products and provided a clearer basis for product-mix, inventory, and market-performance discussions.
- **Key Skills:** Excel analysis, data cleaning, KPI development, PivotTables, interactive dashboards, product-performance analysis, geographic analysis, business interpretation, and data storytelling.

Education

MBA (GPA: 4.0/4.0) | University of Maryland Global Campus
Relevant Coursework: Marketing Management

Expected May 2028

M.S., Data Analytics (GPA: 4.0/4.0) | University of Maryland Global Campus
Relevant Coursework: Machine Learning, Data Management, Data Visualization, Decision Analytics, and AI Ethics

Expected May 2027

B.A., Chemistry | University of Maryland Baltimore County

Dec 2020